EXPERIMENT - 8

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Partition Array for maximum Sum. Leetcode 1043.

Ans :- dp[i] record maximum sum we can get considering A[0] ~ A[i-1].

To get dp[i], we will try to change k last numbers seperately to the maximum of them, where k=1 to k=K.

```
N = A.length,   dp[] = new int[N+1].
for (i=1; i<=N; i++) {
    int currMax=0, best = 0;
    for (int k=1; k<=K && i-k>=0; ++k){
        currMax = Math.max (currMax, A[i-k]);
        best = Math.max( best, dp[i-k] + currMax*k);
    }
    dp[i] = best;
}
return dp[N];
```